Equity Production Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Production Intensity (EPI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on EPI achieves an annualized gross (net) Sharpe ratio of 0.25 (0.23), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of -18 (12) bps/month with a t-statistic of -2.18 (1.50), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Operating leverage, Sales-to-price, Cash-flow to price variance, Advertising Expense, Order backlog, Cash to assets) is -21 bps/month with a t-statistic of -3.56.

1 Introduction

Production-based asset pricing theory fundamentally links firm investment decisions and production technologies to the cross-section of expected returns through the first-order conditions of firm optimization. While traditional asset pricing models focus on consumption-based risk factors, the production approach offers a complementary perspective by examining how firms' optimal investment policies and production constraints generate variation in risk premia across assets. Despite substantial theoretical development in production-based models following Cochrane (1991) and Zhang (2005), empirical evidence linking production characteristics to expected returns remains incomplete, particularly regarding how production intensity affects firms' exposure to systematic risks and their required returns.

In a production economy, firms with higher equity production intensity face distinct marginal costs of capital and adjustment frictions that fundamentally alter their risk profiles. Following the q-theory framework of Liu et al. (2009), firms' investment decisions reflect the shadow value of installed capital, where production intensity directly affects the marginal product of capital and the severity of adjustment costs. When production requires substantial equity capital relative to other inputs, firms experience heightened exposure to productivity shocks because their operating leverage amplifies the impact of demand fluctuations on profitability Carlson et al. (2004). This mechanism creates systematic variation in conditional risk premia, as equity-intensive producers cannot easily adjust their capital stock in response to aggregate shocks.

The production function's curvature and the nature of adjustment costs jointly determine how equity production intensity translates into expected returns. In models with convex adjustment costs, such as Cooper and Haltiwanger (2006), firms with higher equity intensity face steeper marginal costs when adjusting capacity, creating an asymmetric response to positive and negative productivity shocks. This asymmetry generates time-varying risk premia that increase during periods of heightened uncertainty or negative aggregate productivity growth. Moreover, equity-intensive production technologies often exhibit decreasing returns to scale, implying that these firms' marginal q declines more rapidly with investment, leading to lower average returns in equilibrium Belo et al. (2014).

The interaction between equity production intensity and financial frictions further shapes the cross-sectional distribution of expected returns. Following Gomes and Schmid (2016), firms with production technologies requiring substantial equity investment face binding financial constraints more frequently, particularly during economic downturns when external financing becomes costly. These constraints prevent equity-intensive producers from exploiting positive net present value projects, reducing their exposure to systematic risk factors and lowering their expected returns. The production-based model thus predicts that equity production intensity negatively relates to expected returns through both the direct effect on operating leverage and the indirect effect through financial constraints.

We find that a value-weighted long/short trading strategy based on Equity Production Intensity (EPI) achieves an annualized gross Sharpe ratio of 0.25, with the strategy earning an average monthly return of 0.19% (t-statistic = 1.94). The strategy's performance exhibits substantial variation across different factor model specifications, with alphas ranging from -0.18% to 0.13% per month. Most notably, the strategy generates a monthly alpha of -0.18% (t-statistic = -2.18) relative to the Fama-French six-factor model, indicating that EPI captures risk exposures beyond traditional factor models.

The economic significance of the EPI effect varies considerably across firm size categories, consistent with production-based theories predicting stronger effects where financial frictions bind more tightly. Among the largest stocks (those above the 80th NYSE percentile), the EPI strategy achieves an average return of 15 basis points per month (t-statistic = 1.42), with six-factor alphas of -21 basis points (t-statistic = -2.14). The strategy loads significantly on the SMB factor with a coefficient of 0.44 (t-statistic = 15.38) and the RMW factor with a coefficient of 0.56 (t-statistic = 14.45), confirming that equity production intensity relates systematically to size and profitability characteristics.

Controlling for closely related anomalies substantially attenuates the EPI strategy's abnormal returns, suggesting that equity production intensity partially overlaps with existing production-based predictors. When we control for the six most closely related anomalies—Operating leverage, Sales-to-price, Cash-flow to price variance, Advertising Expense, Order backlog, and Cash to assets—along with the Fama-French six factors, the strategy's alpha becomes -21 basis points per month (t-statistic = -3.56). After accounting for transaction costs using the composite bid-ask spreads from Chen and Velikov (2022), the net Sharpe ratio equals 0.23, placing the strategy in the top 19% of anomalies in terms of implementable performance.

This paper contributes to the production-based asset pricing literature by providing comprehensive evidence on how equity production intensity predicts cross-sectional returns, extending the theoretical frameworks of Zhang (2005) and Liu et al. (2009). Unlike Novy-Marx (2011), who examines operating leverage through the ratio of fixed to variable costs, we focus specifically on the equity intensity of production, capturing a distinct dimension of production risk that reflects both technological constraints and financial frictions. Our results complement Hou et al. (2015) by demonstrating that production characteristics beyond investment and profitability contain incremental information about expected returns, particularly when these characteristics interact with firms' financing constraints.

Methodologically, we advance the literature by employing the comprehensive testing framework of Novy-Marx and Velikov (2023b), which ensures robust inference across multiple portfolio construction methods and accounts for implementation costs that often eliminate apparent anomaly profits. This approach reveals that while EPI generates statistically significant gross returns, its net performance depends critically on portfolio construction methodology and the control for related production-based factors. By examining EPI's performance relative to 212 documented anomalies, we show that production intensity ranks in the top quartile of implementable strategies after transaction costs, suggesting genuine economic content beyond data mining concerns raised by Harvey et al. (2016).

Our findings have broader implications for understanding the sources of cross-sectional return variation and the limits of arbitrage in production economies. The negative alpha relative to standard factor models suggests that equity production intensity proxies for a risk factor that existing models fail to capture fully, potentially related to time-varying adjustment costs or production inflexibility during economic downturns. The strategy's modest performance among large-cap stocks, where production-based theories predict weaker effects due to lower adjustment costs and fewer financial frictions, supports the economic mechanisms underlying production-based asset pricing models. These results underscore the importance of incorporating production characteristics into asset pricing models and suggest that the cross-section of returns reflects fundamental economic forces related to firms' production technologies and investment frictions.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of balance sheet and income statement variables across firms.signal construction relies on two key accounting variables. Common equity (CEQ) represents the total common shareholders' equity as reported on the balance sheet, calculated as total shareholders' equity minus the liquidation value of preferred stock. This measure captures the book value of equity attributable to common shareholders, reflecting the cumulative retained earnings and contributed capital net of distributions. Cost of goods sold (COGS) represents the direct costs attributable to the production of goods sold by a company during the fiscal year, including raw materials, direct labor, and manufacturing overhead. This income statement item measures the resources consumed in the production process during the reporting period.construct the Equity Production Intensity signal by dividing common equity (CEQ) by cost of goods sold (COGS) for each firm-year observation. This ratio captures the relationship between a firm's equity base and its production activity, measuring how much shareholder equity supports each dollar of production costs. A higher ratio indicates greater equity intensity relative to production expenses, potentially reflecting differences in capital structure, operational efficiency, or business model characteristics across firms. We calculate this measure using fiscal year-end values and require non-missing, positive values for both CEQ and COGS to ensure meaningful ratios. Firms with zero or negative COGS, which may occur in certain service industries or during unusual reporting periods, are excluded from our analysis.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EPI signal. Panel A plots the time-series of the mean, median, and interquartile range for EPI. On average, the cross-sectional mean (median) EPI is -5.42 (-0.85) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EPI data. The signal's interquartile range spans -4.34 to -0.30. Panel B of Figure 1 plots the time-series of the coverage of the EPI signal for the CRSP universe. On average, the EPI signal is available for 6.93% of CRSP names, which on average make up 8.04% of total market capitalization.

4 Does EPI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EPI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EPI portfolio and sells the low EPI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EPI strategy earns an average return of 0.19% per month with a t-statistic of 1.94. The annualized Sharpe ratio of the strategy is 0.25. The alphas range from -0.18% to 0.13% per month and have t-statistics exceeding -2.18 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.44, with a t-statistic of 15.38 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 595 stocks and an average market capitalization of at least $1,321 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 19 bps/month with a t-statistics of 1.94. Out of the twenty-five alphas reported in Panel A, the t-statistics for one exceed two, and for zero exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 6-28bps/month. The lowest return, (6 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.65. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EPI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in one cases.

Table 3 provides direct tests for the role size plays in the EPI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EPI, as well as average returns and alphas for long/short trading EPI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EPI strategy achieves an average return of 15 bps/month with a t-statistic of 1.42. Among these large cap stocks, the alphas for the EPI strategy relative to the five most common factor models range from -21 to 11 bps/month with t-statistics between -2.14 and 1.06.

5 How does EPI perform relative to the zoo?

Figure 2 puts the performance of EPI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EPI strategy falls in the distribution. The EPI strategy's gross (net) Sharpe ratio of 0.25 (0.23) is greater than 51% (81%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EPI strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EPI strategy

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that

would have yielded \$1.97 which ranks the EPI strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EPI strategy would have yielded \$1.64 which ranks the EPI strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EPI relative to those. Panel A shows that the EPI strategy gross alphas fall between the 6 and 32 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EPI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EPI ranks between the 50 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EPI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EPI with 210 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EPI or at least to weaken the power EPI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI} EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EPI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EPI signal in these Fama-MacBeth regressions exceed

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-1.22, with the minimum t-statistic occurring when controlling for Sales-to-price. Controlling for all six closely related anomalies, the t-statistic on EPI is -0.66.

Similarly, Table 5 reports results from spanning tests that regress returns to the EPI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EPI strategy earns alphas that range from -27–10bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -4.00, which is achieved when controlling for Sales-to-price. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EPI trading strategy achieves an alpha of -21bps/month with a t-statistic of -3.56.

7 Does EPI add relative to the whole zoo?

Finally, we can ask how much adding EPI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EPI signal.[4] We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes EPI grows to $3165.09.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EPI is available.

8 Conclusion

This study provides a comprehensive evaluation of Equity Production Intensity (EPI) as a predictor of cross-sectional stock returns using the rigorous testing framework of Novy-Marx and Velikov (2023). Our findings reveal that while EPI demonstrates some predictive power with a modest Sharpe ratio of 0.25 (0.23 net of costs), its economic significance appears limited when subjected to more stringent tests. The strategy generates negative abnormal returns of -18 basis points per month relative to the six-factor model, though this reverses to a positive but statistically insignificant 12 basis points after accounting for transaction costs. Most notably, when controlling for closely related factors including operating leverage, sales-to-price, and other fundamental metrics, the strategy produces a significantly negative alpha of -21 basis points per month (t-statistic = -3.56), suggesting that EPI's apparent predictive power is largely subsumed by existing factors in the literature.

These results have important implications for both academic research and investment practice. For researchers, our findings underscore the importance of rigorous testing protocols that account for multiple comparison issues and the proliferation of factors in the asset pricing literature. For practitioners, while EPI may offer some diversification benefits, its standalone implementation appears unlikely to generate significant risk-adjusted returns after accounting for implementation costs and exposure to known risk factors.

Several limitations of this study warrant consideration. First, our analysis is confined to U.S. equity markets, and the performance of EPI in international markets remains unexplored. Second, the sample period and specific implementation choices may influence the results, and robustness across different market regimes deserves further investigation. Third, while we control for several related factors, the interaction effects between EPI and other firm characteristics in different economic conditions merit additional analysis.

Future research should explore several promising avenues. Investigation of EPI's performance in international markets could reveal whether its predictive power varies across different institutional and regulatory environments. Additionally, examining the time-varying nature of EPI's effectiveness, particularly during periods of technological disruption or economic stress, could provide insights into when this signal might be most valuable. Researchers might also investigate whether combining EPI with machine learning techniques or alternative data sources could enhance its predictive power. Finally, a deeper theoretical exploration of why EPI's information content appears to be captured by existing factors would contribute to our understanding of the underlying economic mechanisms driving cross-sectional return variation.

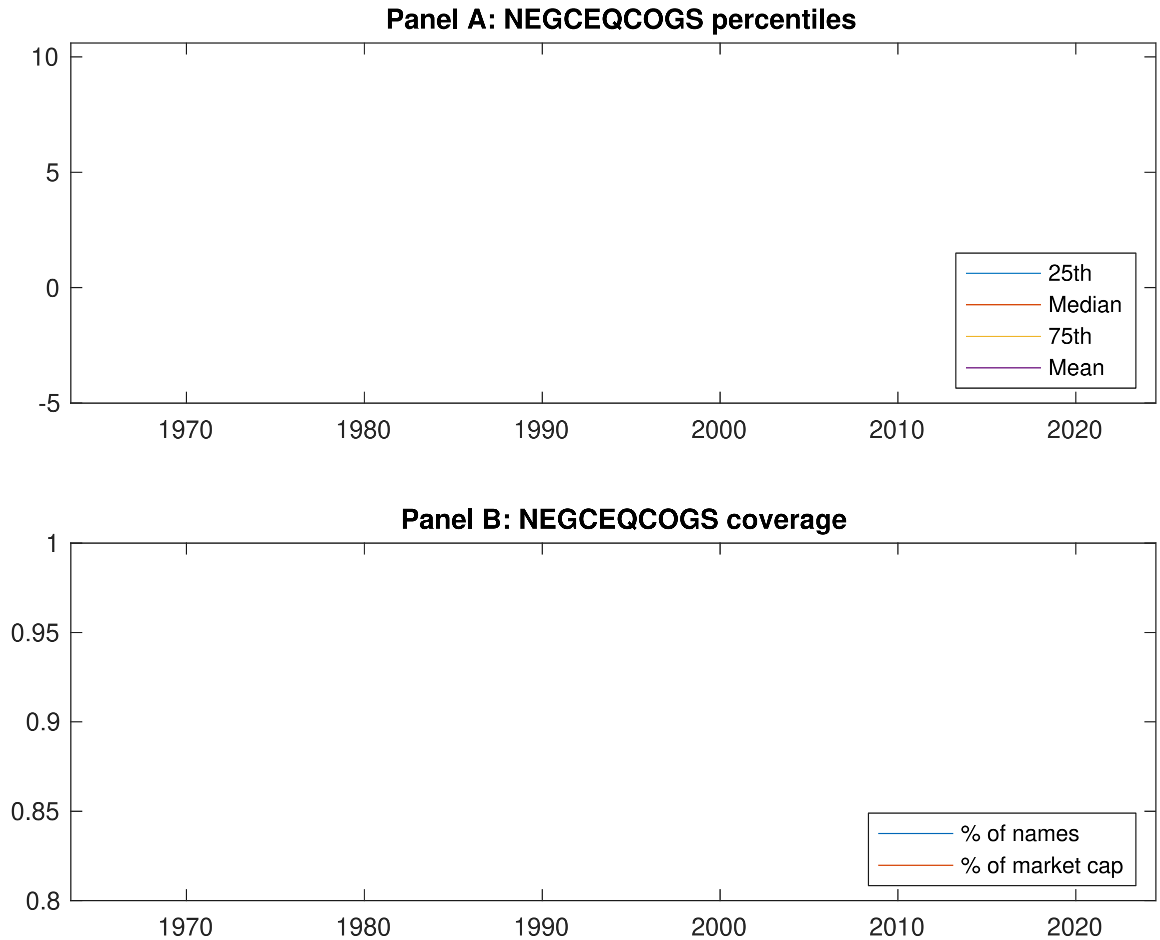


Figure 1: Times series of EPI percentiles and coverage.
This figure plots descriptive statistics for EPI. Panel A shows cross-sectional percentiles of EPI over the sample. Panel B plots the monthly coverage of EPI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EPI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EPI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54	0.56	0.62	0.67	0.72	0.19
	[3.27]	[3.43]	[3.56]	[3.74]	[3.91]	[1.94]
α_{CAPM}	-0.03	-0.00	0.02	0.07	0.11	0.13
	[-0.57]	[-0.11]	[0.48]	[1.13]	[1.63]	[1.37]
α_{FF3}	0.01	-0.02	-0.01	0.03	0.06	0.05
	[0.27]	[-0.40]	[-0.24]	[0.48]	[0.95]	[0.52]
α_{FF4}	0.03	-0.01	0.01	0.04	0.05	0.02
	[0.71]	[-0.35]	[0.26]	[0.63]	[0.73]	[0.17]
α_{FF5}	0.12	-0.05	-0.11	-0.08	-0.05	-0.17
	[3.20]	[-1.30]	[-2.39]	[-1.37]	[-0.80]	[-2.09]
α_{FF6}	0.13	-0.05	-0.08	-0.06	-0.05	-0.18
	[3.32]	[-1.15]	[-1.76]	[-1.04]	[-0.85]	[-2.18]
Panel B: Fama and French (2018) 6-factor model loadings for EPI-sorted portfolios						
β_{MKT}	0.94	1.00	1.04	1.04	1.04	0.10
	[100.07]	[106.05]	[95.73]	[76.11]	[73.89]	[5.13]
β_{SMB}	-0.17	-0.09	0.08	0.11	0.27	0.44
	[-12.72]	[-6.57]	[4.86]	[5.82]	[13.10]	[15.38]
β_{HML}	0.01	-0.00	0.03	0.00	0.07	0.06
	[0.58]	[-0.20]	[1.39]	[0.19]	[2.70]	[1.65]
β_{RMW}	-0.24	0.03	0.25	0.24	0.32	0.56
	[-12.92]	[1.58]	[11.69]	[8.85]	[11.67]	[14.45]
β_{CMA}	-0.17	0.13	0.09	0.15	-0.01	0.16
	[-6.36]	[4.83]	[2.86]	[3.86]	[-0.27]	[2.83]
β_{UMD}	-0.01	-0.01	-0.04	-0.03	0.01	0.01
	[-0.98]	[-0.80]	[-3.72]	[-1.87]	[0.37]	[0.73]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1140	676	595	597	714	
me ($\$10^6$)	3543	2221	1557	1651	1321	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EPI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.19	0.13	0.05	0.02	-0.17	-0.18
			[1.94]	[1.37]	[0.52]	[0.17]	[-2.09]	[-2.18]
Quintile	NYSE	EW	0.19	0.10	-0.00	0.04	-0.13	-0.09
			[1.96]	[1.04]	[-0.01]	[0.42]	[-1.59]	[-1.05]
Quintile	Name	VW	0.30	0.25	0.17	0.11	-0.12	-0.14
			[2.76]	[2.34]	[1.64]	[1.07]	[-1.33]	[-1.60]
Quintile	Cap	VW	0.22	0.19	0.10	0.05	-0.19	-0.20
			[2.20]	[1.83]	[1.00]	[0.51]	[-2.15]	[-2.30]
Decile	NYSE	VW	0.26	0.19	0.11	0.05	-0.14	-0.17
			[2.32]	[1.72]	[1.05]	[0.47]	[-1.47]	[-1.74]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.17	0.11	0.04	0.02	0.12	0.12
			[1.77]	[1.16]	[0.46]	[0.25]	[1.42]	[1.50]
Quintile	NYSE	EW	0.06					
			[0.65]					
Quintile	Name	VW	0.28	0.23	0.16	0.12	0.05	0.07
			[2.59]	[2.11]	[1.54]	[1.20]	[0.56]	[0.75]
Quintile	Cap	VW	0.21	0.16	0.09	0.06	0.13	0.14
			[2.06]	[1.59]	[0.89]	[0.60]	[1.45]	[1.58]
Decile	NYSE	VW	0.24	0.17	0.10	0.06	0.06	0.08
			[2.12]	[1.49]	[0.96]	[0.62]	[0.67]	[0.86]

Table 3: Conditional sort on size and EPI

This table presents results for conditional double sorts on size and EPI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EPI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EPI and short stocks with low EPI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

	EPI Quintiles					EPI Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.51	0.76	0.81	0.83	0.87	0.36	0.26	0.18	0.23	0.05	0.11
	[2.21]	[3.17]	[3.33]	[3.39]	[3.15]	[2.64]	[1.95]	[1.33]	[1.72]	[0.39]	[0.84]
(2)	0.58	0.86	0.83	0.74	0.89	0.31	0.23	0.13	0.18	-0.17	-0.10
	[2.57]	[3.91]	[3.71]	[3.25]	[3.64]	[2.56]	[1.96]	[1.10]	[1.48]	[-1.57]	[-0.94]
(3)	0.51	0.67	0.77	0.82	0.93	0.42	0.35	0.23	0.23	-0.12	-0.09
	[2.42]	[3.40]	[3.78]	[3.89]	[4.12]	[3.32]	[2.81]	[1.86]	[1.85]	[-1.10]	[-0.80]
(4)	0.62	0.63	0.75	0.72	0.75	0.13	0.12	0.00	-0.01	-0.37	-0.35
	[3.00]	[3.50]	[3.95]	[3.56]	[3.67]	[0.99]	[0.89]	[0.01]	[-0.05]	[-3.46]	[-3.21]
(5)	0.47	0.52	0.53	0.64	0.62	0.15	0.11	0.04	-0.01	-0.18	-0.21
	[2.82]	[3.16]	[3.26]	[3.81]	[3.53]	[1.42]	[1.06]	[0.40]	[-0.09]	[-1.91]	[-2.14]

Panel B: Portfolio average number of firms and market capitalization

	EPI Quintiles: Average n					EPI Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	418	418	418	416	402	36	36	35	35	31
(2)	116	116	116	116	116	58	60	60	59	59
(3)	83	83	83	83	83	101	102	103	104	104
(4)	68	69	69	69	69	214	217	221	216	220
(5)	62	62	63	62	62	1861	1832	1438	1670	1421

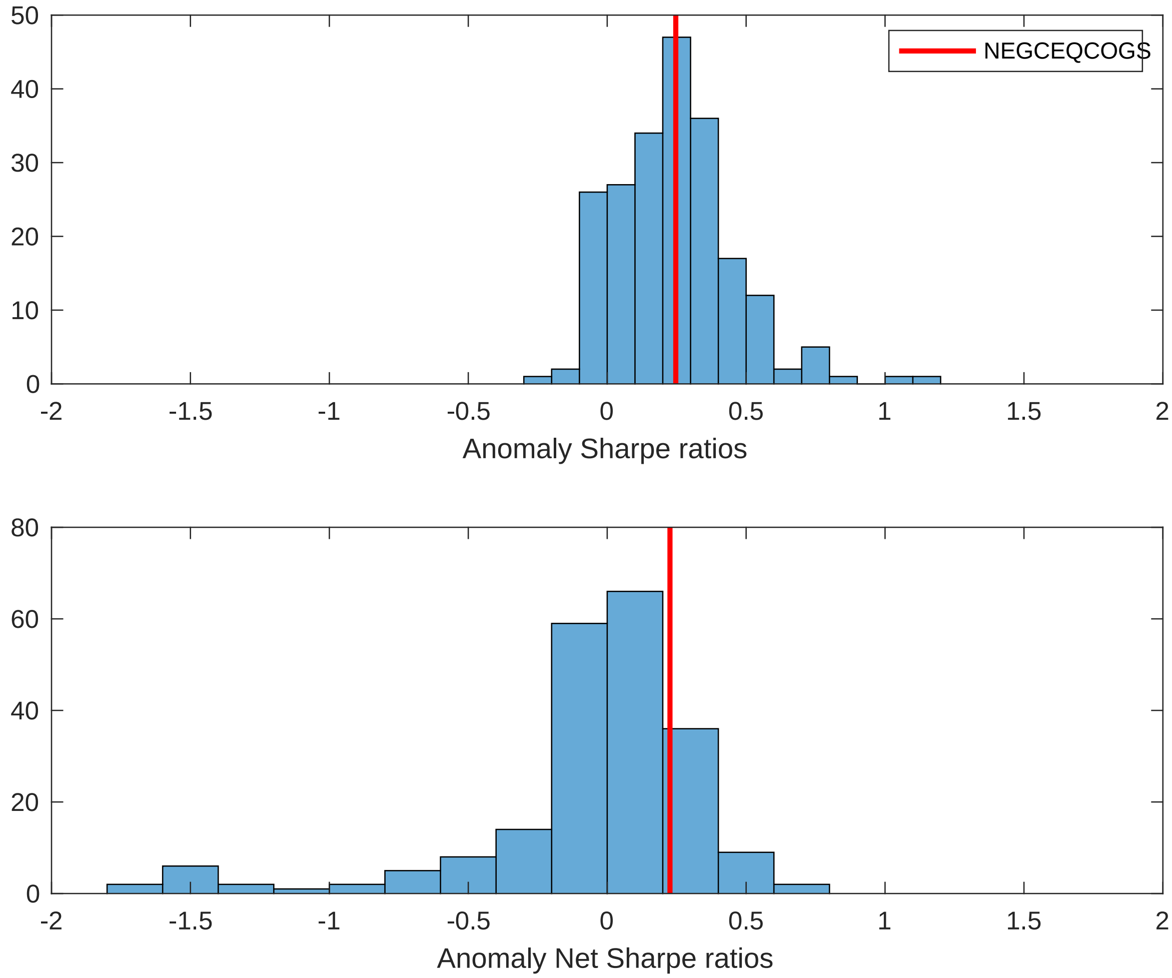


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EPI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

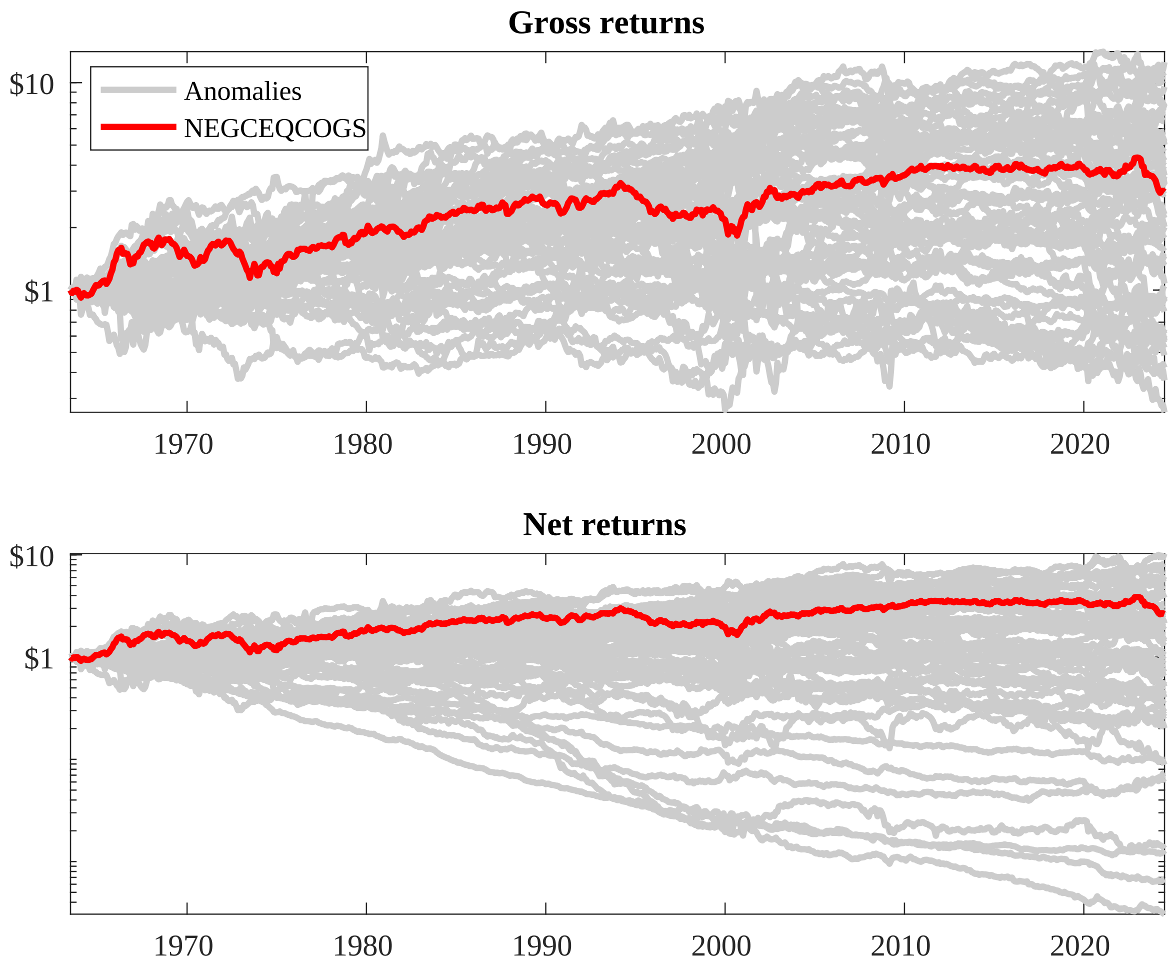
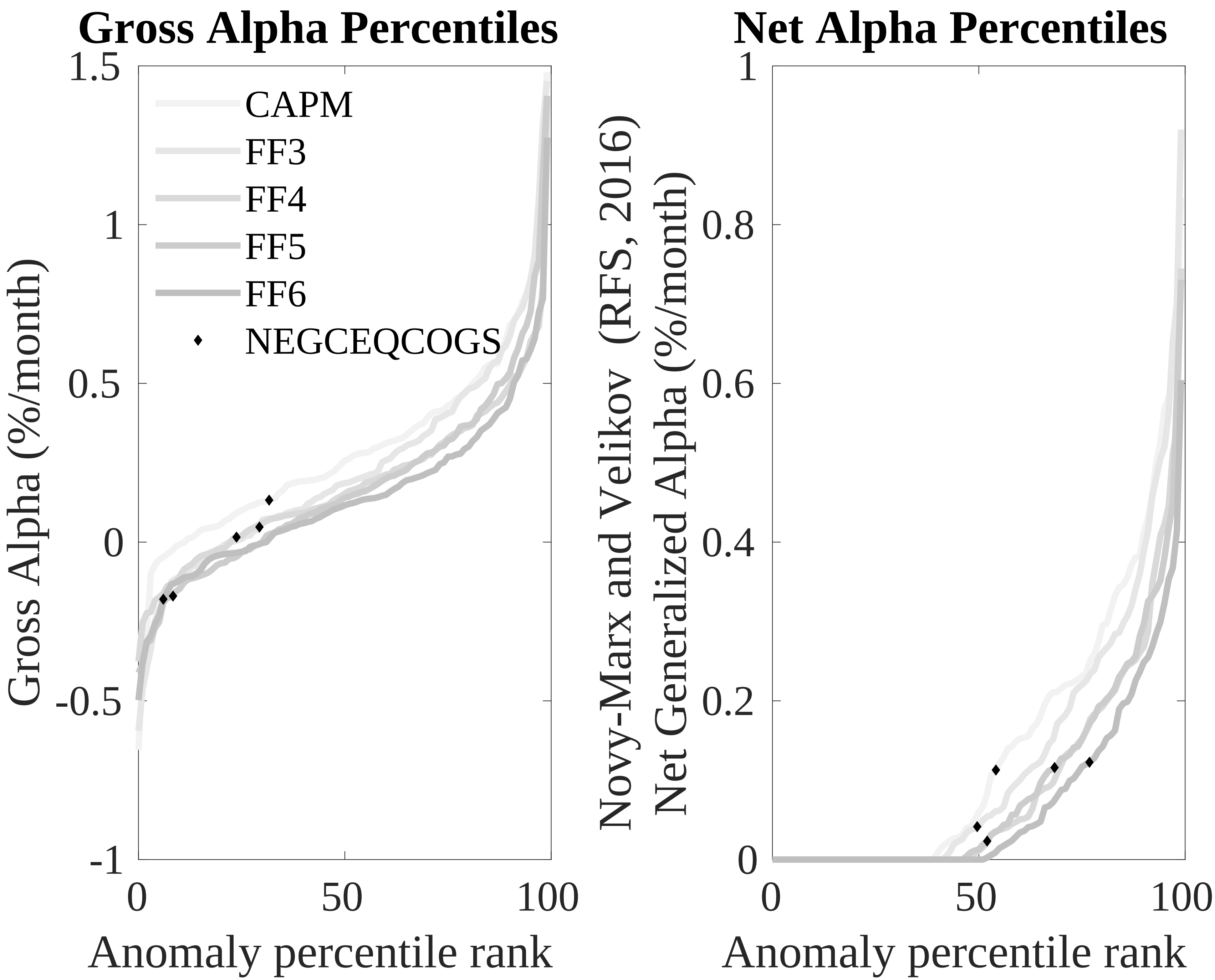


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EPI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGCEQCOGS
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

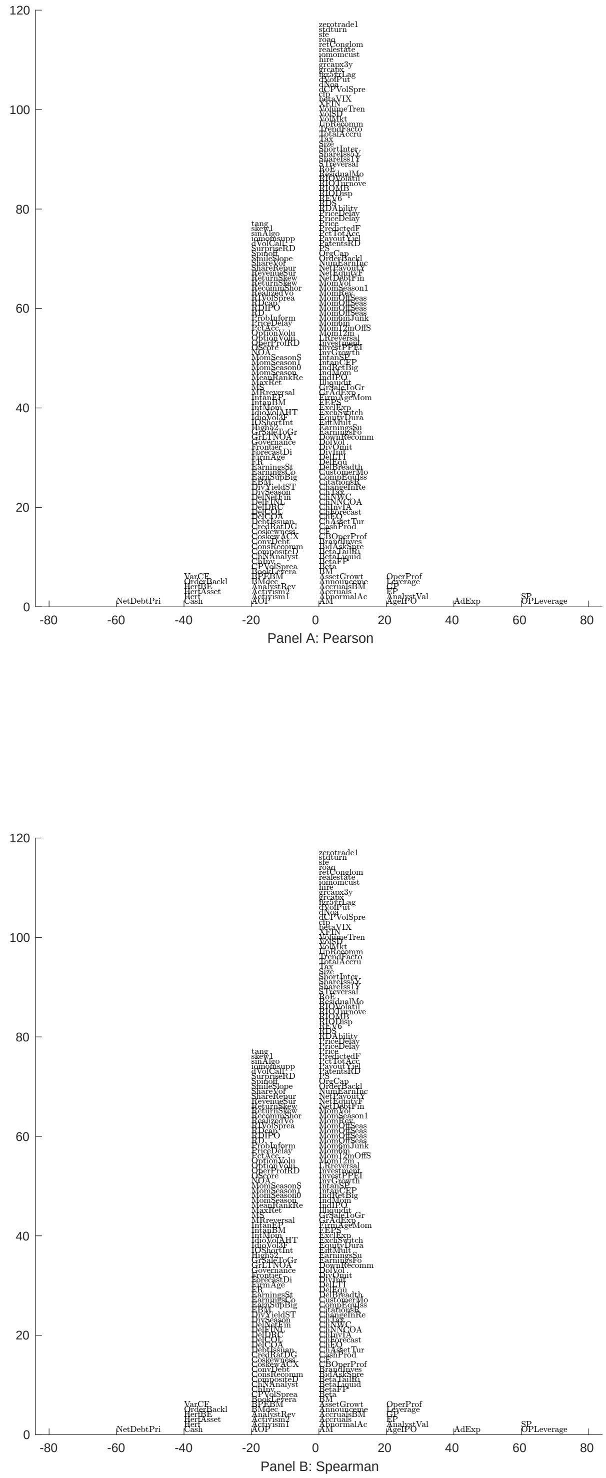


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EPI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

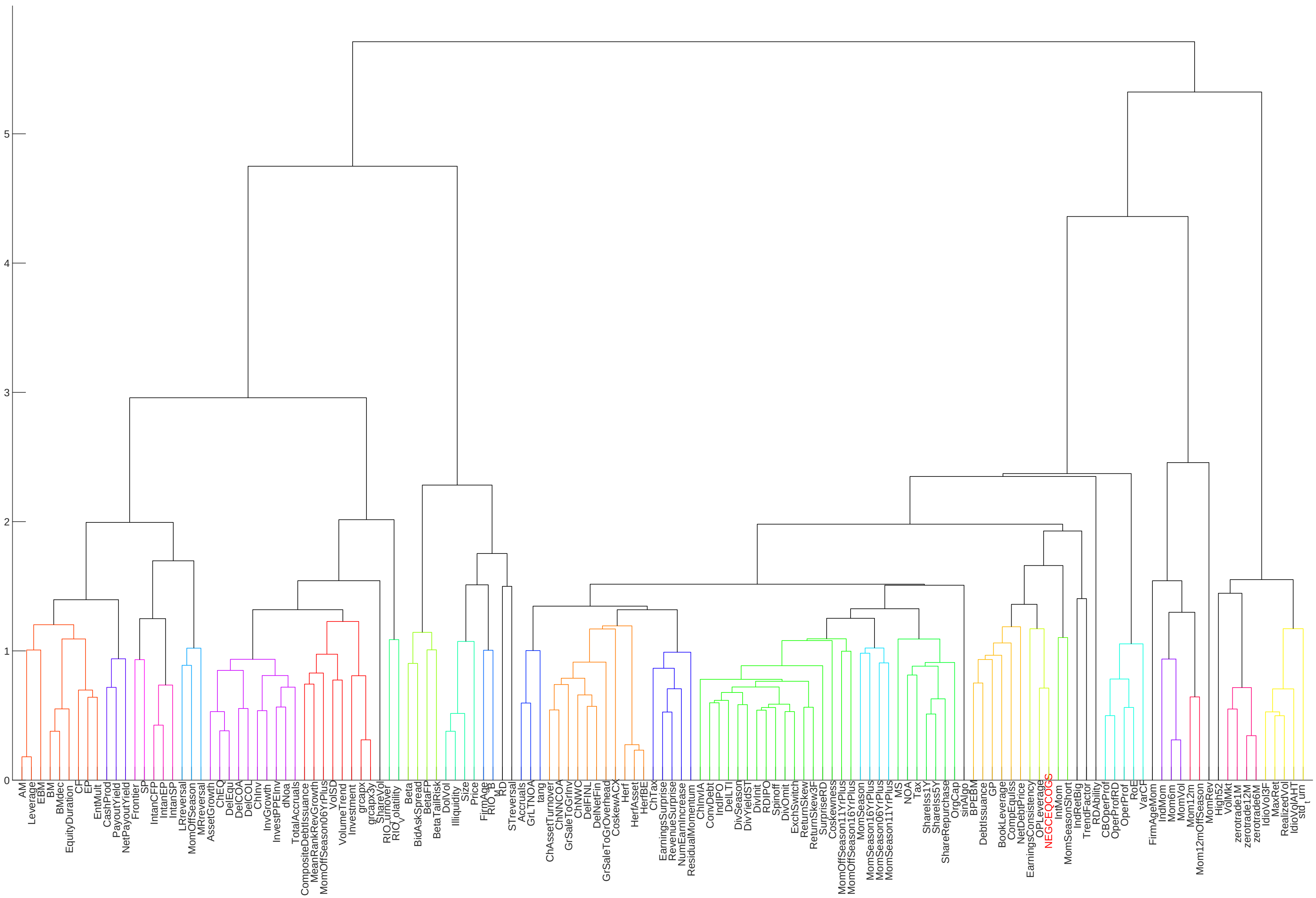


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

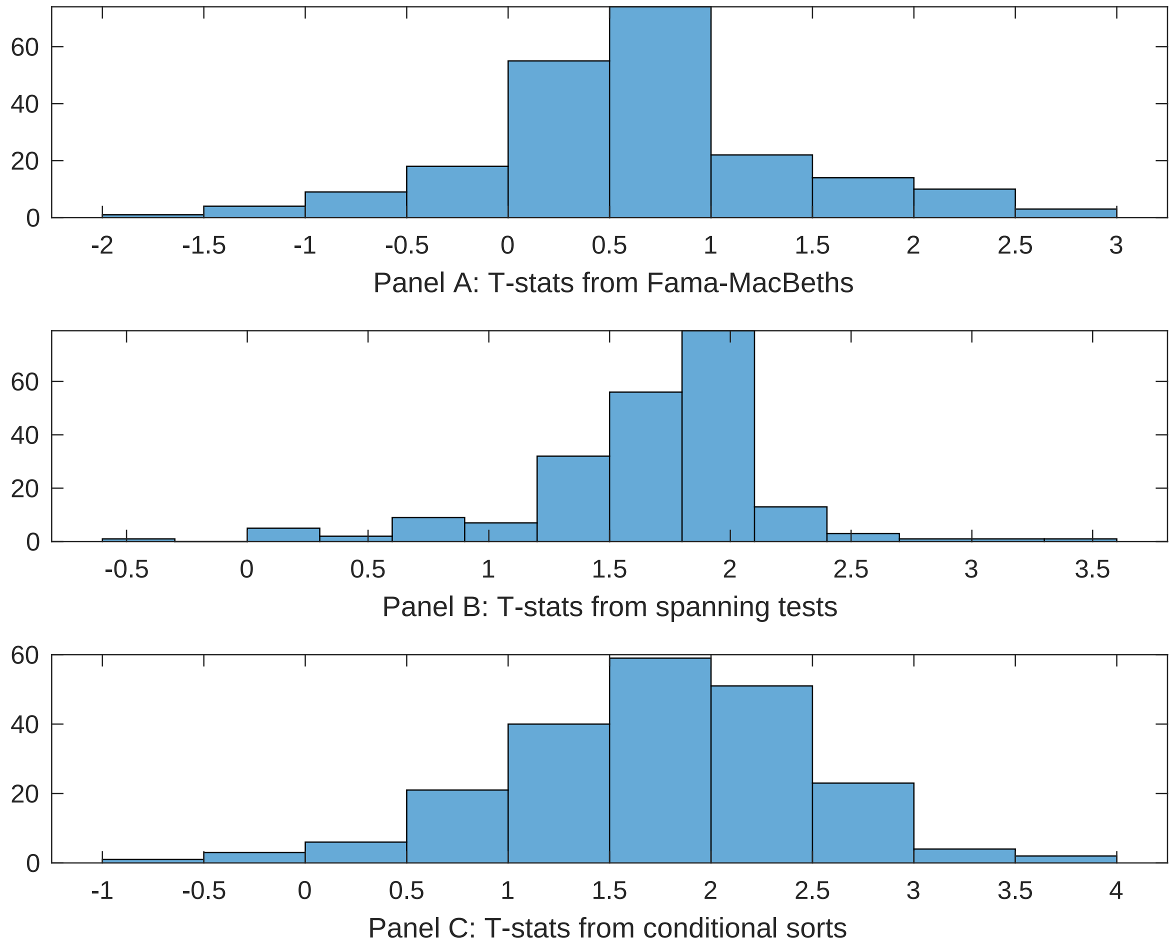


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI} EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EPI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EPI} EPI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Operating leverage, Sales-to-price, Cash-flow to price variance, Advertising Expense, Order backlog, Cash to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.10	0.98	0.12	0.11	0.14	0.12	0.76
	[5.00]	[4.75]	[5.74]	[4.69]	[5.14]	[5.01]	[2.33]
EPI	-0.59	-0.13	0.53	-0.16	0.76	0.45	-0.54
	[-0.53]	[-1.22]	[0.41]	[-1.13]	[2.52]	[2.40]	[-0.66]
Anomaly 1	0.13						0.87
	[2.88]						[0.58]
Anomaly 2		0.69					0.96
		[4.09]					[2.27]
Anomaly 3			-0.37				0.12
			[-1.88]				[1.54]
Anomaly 4				0.26			-0.69
				[0.52]			[-0.53]
Anomaly 5					0.71		0.17
					[0.02]		[0.11]
Anomaly 6						0.87	0.19
						[2.47]	[4.16]
# months	732	727	727	619	636	626	582
$\bar{R}^2$(%)	0	1	1	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EPI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EPI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Operating leverage, Sales-to-price, Cash-flow to price variance, Advertising Expense, Order backlog, Cash to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	-0.27	-0.25	-0.16	-0.20	-0.17	-0.10	-0.21
	[-4.00]	[-3.74]	[-1.98]	[-2.38]	[-2.00]	[-1.15]	[-3.56]
Anomaly 1	53.43						43.85
	[19.89]						[17.69]
Anomaly 2		53.27					27.00
		[18.82]					[9.03]
Anomaly 3			-22.47				-3.66
			[-7.30]				[-1.47]
Anomaly 4				11.11			4.32
				[5.19]			[2.34]
Anomaly 5					-12.12		-10.17
					[-6.16]		[-7.40]
Anomaly 6						-20.36	-17.42
						[-6.26]	[-7.56]
mkt	10.58	3.30	3.89	9.90	9.46	12.66	8.82
	[6.62]	[1.98]	[1.85]	[5.06]	[4.81]	[6.22]	[5.87]
smb	20.11	10.86	27.86	39.82	35.43	37.99	-3.42
	[7.72]	[3.69]	[7.82]	[13.53]	[11.89]	[12.90]	[-1.25]
hml	34.86	-25.83	-2.99	1.62	0.70	-4.33	1.22
	[10.26]	[-7.18]	[-0.76]	[0.42]	[0.19]	[-1.10]	[0.36]
rmw	26.56	41.61	63.17	54.03	48.01	45.52	11.12
	[7.70]	[12.74]	[16.34]	[14.10]	[12.14]	[11.17]	[3.42]
cma	5.16	1.54	11.05	10.52	25.01	16.26	-4.49
	[1.13]	[0.33]	[2.02]	[1.86]	[4.36]	[2.78]	[-1.11]
umd	0.30	22.95	4.57	5.39	-2.31	-0.20	12.22
	[0.19]	[11.61]	[2.35]	[2.59]	[-1.18]	[-0.10]	[7.16]
# months	732	728	728	716	642	629	629
$\bar{R}^2(\%)$	58	57	40	38	37	38	72

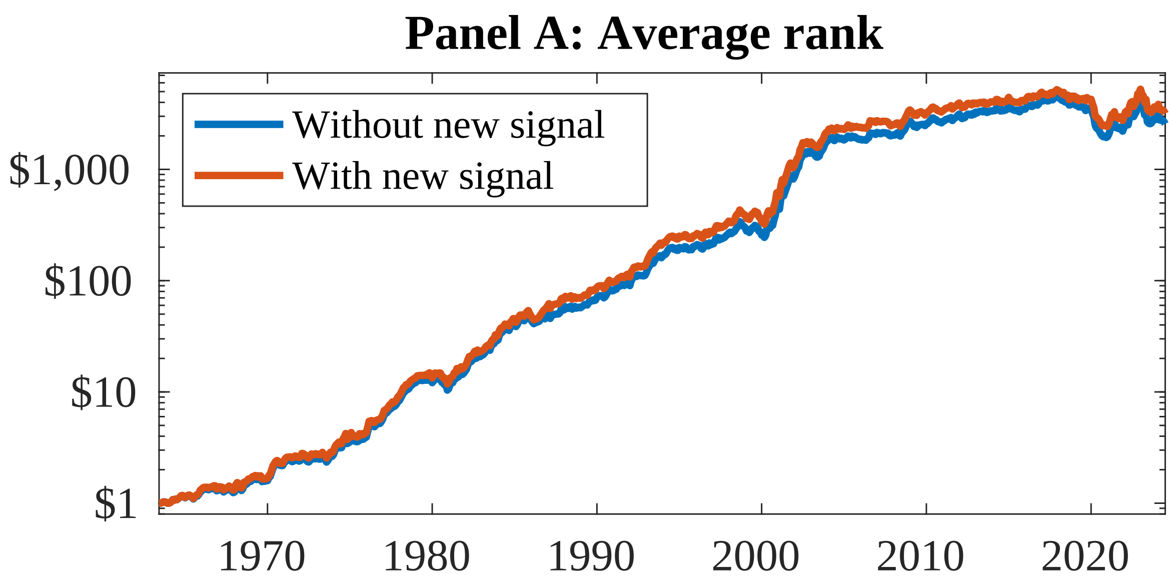


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EPI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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